

Prompt Engineering Handbook

Designing Reliable Instructions for Generative AI

BUILT FOR REAL WORK

A practical handbook for professionals, students, analysts, builders, career switchers, and anyone who wants more reliable results from modern AI systems.

10

Core chapters

60+

Prompt patterns

25+

Worked examples

1

Capstone workflow

CareerSense

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FRONT MATTER

How to Use This Handbook

Prompt engineering is not about memorizing magic words. It is about reducing ambiguity, supplying useful context, and designing a repeatable way to get the kind of output you actually need.

Every chapter follows the same learning rhythm:

1. Start with the core idea and why it matters in real work.
2. See the pattern in a small prompt before using it in a larger workflow.
3. Compare weak and improved prompts side by side.
4. Study common failure modes and how to repair them.
5. Finish with quick revision notes, practice questions, and hands-on exercises.

IMPORTANT

AI systems can still make mistakes. Treat generated output as a draft or assistant contribution, especially for high-stakes decisions, confidential information, legal, financial, medical, or security-sensitive work.

Use this book to

- Write clearer prompts for everyday work
- Build reusable templates
- Evaluate outputs consistently
- Create multi-step AI workflows

Do not use it to

- Hide critical decisions behind AI
- Paste sensitive data without approval
- Assume fluent output is always correct
- Replace domain judgment

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What You Will Learn

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CHAPTER 01

How Generative AI Interprets Instructions

Before you improve prompts, understand what the model is actually responding to.

Learning objectives

- Explain why prompts are instructions plus context, not commands with guaranteed outcomes
- Distinguish intent, context, constraints, and examples
- Recognize ambiguity and underspecified tasks
- Understand why wording, ordering, and available context affect output

The model is predicting a useful continuation

A generative model does not read your mind. It receives the conversation and produces a response that is statistically compatible with the instructions, examples, context, and patterns it has learned. Clear prompts reduce the number of plausible but unwanted interpretations.

ANALOGY

Think of prompting like delegating work to a capable new teammate. “Make this better” forces them to guess. “Rewrite this for a hiring manager, keep it under 120 words, preserve the numbers, and end with a clear next step” gives them a usable brief.

Weak prompt

- Summarize this
- Make it professional
- Analyze the data
- Write an email

Better prompt

- Summarize for a VP in 5 bullets
- Rewrite for a formal customer email
- Find 3 patterns, 2 risks, and 1 next action
- Draft a 100-word follow-up with a polite CTA

BEST PRACTICE

Before sending a prompt, ask: “What would another person still have to guess?” Add only the missing information that materially changes the result.

CHAPTER 02

Anatomy of a Strong Prompt

Reliable prompts usually have a small number of repeatable building blocks.

Learning objectives

- Write a clear task statement
- Provide only relevant context
- Add constraints that reduce ambiguity
- Specify an output format
- Define quality criteria

Component	Question it answers	Example
Task	What should the model do?	Compare these two job offers.
Context	What should it know?	I value learning and work-life balance.
Constraints	What must it avoid or obey?	Do not assume relocation is possible.
Output	What shape should the answer take?	Use a 5-row comparison table.
Quality	What does good look like?	Highlight trade-offs and uncertainties.

PROMPT TEMPLATE

Task: [what you want done]

Context: [relevant facts]

Constraints: [limits, must-haves, must-avoid]

Output: [format, length, structure]

Quality bar: [criteria for a good answer]

WORKED EXAMPLE

Task: Rewrite my project description for a data analyst resume.

Context: I automated a weekly report using SQL and Power BI. It reduced preparation time from 5 hours to 2 hours.

Constraints: Do not invent metrics or tools. Keep it truthful.

Output: 3 bullet options, each under 28 words.

Quality bar: Start with a strong action verb and emphasize measurable impact.

COMMON MISTAKE

Adding a long persona while leaving the actual task vague. A precise task usually matters more than decorative role-play.

CHAPTER 03

Context, Constraints, and Output Formats

Most prompt quality problems are really specification problems.

Learning objectives

- Separate useful context from distracting context
- Use positive and negative constraints
- Control length and structure
- Use schemas, tables, and checklists when consistency matters

Context should change the answer

Useful context narrows the problem. If removing a detail would not materially change the desired response, it probably does not belong in the prompt.

Useful context

- Audience and seniority
- Goal and deadline
- Source material
- Known constraints
- Decision criteria

Usually noise

- Long biographies unrelated to task
- Repeated instructions
- Decorative adjectives
- Conflicting constraints
- Unneeded background history

STRUCTURED OUTPUT PROMPT

Analyze the feedback below.

Return JSON with exactly these keys:

```
{
  "theme": "...",
  "severity": "low | medium | high",
  "evidence": ["..."],
  "recommended_action": "..."
}
```

Do not add keys. If evidence is insufficient, set severity to "unknown".

BEST PRACTICE

When output will be consumed by code, specify a schema. When output will be consumed by a person, specify headings, bullets, or a table.

CHAPTER 04

Examples and Few-Shot Prompting

Sometimes the fastest way to explain quality is to show it.

Learning objectives

- Use examples to demonstrate tone, labels, or structure
- Choose representative examples
- Avoid examples that accidentally encode the wrong rule
- Know when zero-shot instructions are enough

CLASSIFICATION EXAMPLE

Classify each message as Billing, Technical, Account, or Other.

Examples:

"I was charged twice" -> Billing

"The app crashes after login" -> Technical

"I need to change my email" -> Account

Message: "My invoice has the wrong tax amount"

Return only the label.

Good examples teach the boundary

A useful few-shot set includes cases that differ in meaningful ways. If all examples are nearly identical, the model may imitate superficial wording rather than learn the intended distinction.

COMMON MISTAKE

Giving examples that contradict the written instruction. The model may follow the demonstrated pattern instead of the sentence you intended.

- Use examples when the output style is difficult to describe.
- Use examples when categories are subtle or business-specific.
- Avoid unnecessary examples for simple, well-specified tasks.
- Keep labels and formats consistent across examples.

CHAPTER 05

Reasoning Tasks and Decomposition

Complex tasks are easier when you separate the work into explicit stages.

Learning objectives

- Break broad tasks into smaller deliverables
- Ask for assumptions and missing inputs
- Use checklists for multi-step analysis
- Separate generation from evaluation

DECOMPOSITION PROMPT

Help me choose between two career options.

Step 1: List the decision criteria explicitly stated in my notes.

Step 2: Identify missing information that could materially change the decision.

Step 3: Compare the options against each criterion.

Step 4: Give a recommendation with uncertainty and trade-offs.

Do not invent salary, visa, or company facts.

PROFESSIONAL PATTERN

Generate -> Check -> Improve is often more reliable than asking for a perfect answer in one pass. First create the draft, then evaluate against criteria, then revise only the weak parts.

When to decompose

- Research synthesis
- Planning
- Debugging
- Long-form writing
- Multi-criteria decisions

Keep it simple for

- Short rewrites
- Simple extraction
- Basic formatting
- Direct factual transformations

CHAPTER 06

Roles, Personas, and Audience Control

Roles can focus vocabulary and perspective, but they should support the task, not replace it.

Learning objectives

- Use roles to set domain perspective
- Specify the real audience separately
- Control tone without vague adjectives
- Avoid false authority

ROLE + AUDIENCE

Act as a senior analytics manager reviewing a junior analyst's project summary.

Audience: a hiring manager who is not deeply technical.

Task: rewrite the summary so the business impact is clear.

Constraints: preserve all facts; remove unnecessary jargon.

Output: 2 versions, each 60-80 words.

IMPORTANT

A persona does not make the model a licensed professional or create verified expertise. Use domain review for high-stakes outputs.

Vague style word	More actionable instruction
Professional	Use concise business language; no slang; one idea per sentence.
Friendly	Warm opening, plain language, no exaggerated enthusiasm.
Executive	Lead with decision, then 3 supporting facts, then risk.
Technical	Use precise terms and include assumptions and edge cases.

CHAPTER 07

Prompting for Data, Code, and Analysis

Technical prompts improve when inputs, assumptions, and validation steps are explicit.

Learning objectives

- Ask for code with environment and constraints
- Request explanations at the right level
- Use data dictionaries and schemas
- Require validation and edge-case handling

CODE PROMPT

Write a Python function that reads a CSV of job applications.

Environment: Python 3.11, pandas allowed.

Columns: company, role, applied_date, status.

Requirements:

- Return monthly application counts by status.
- Handle missing status as "Unknown".
- Do not modify the source file.
- Include one small test example.

Explain the code in 5 bullets after the code block.

ANALYSIS PROMPT

You are given a table of interview outcomes.

Before analyzing:

1. State the grain of the data.
2. List any fields that could cause double-counting.
3. Identify missing or ambiguous definitions.

Then calculate only the metrics supported by the data.

Separate observations from hypotheses.

BEST PRACTICE

For code, specify language, version, libraries, input shape, output shape, constraints, and test expectations. For data, specify grain, definitions, and what must not be inferred.

CHAPTER 08

Prompt Evaluation and Iteration

A prompt is only good if the output consistently meets the need.

Learning objectives

- Define evaluation criteria before tweaking prompts
- Build small test sets
- Distinguish prompt failure from missing data
- Compare versions systematically

Criterion	Question
Accuracy	Are claims supported by the provided information?
Completeness	Did it address every required part?
Relevance	Did it stay focused on the task?
Format	Did it follow the requested structure?
Tone	Is the language right for the audience?
Safety	Did it avoid unsupported or sensitive actions?

EVALUATOR PROMPT

Evaluate the draft against these criteria: Accuracy, Completeness, Relevance, Format, Tone.
 Score each from 1-5.
 For any score below 4, quote the exact weak section and propose one targeted fix.
 Do not rewrite the entire draft unless requested.

ITERATION RULE

Change one major variable at a time. If you simultaneously change the task, examples, output format, and tone, you will not know what actually improved the result.

CHAPTER 09

Reusable Prompt Systems and Workflows

Professional prompt engineering is less about one clever prompt and more about repeatable systems.

Learning objectives

- Turn successful prompts into templates
- Separate stable instructions from variable inputs
- Design multi-step workflows
- Use checkpoints and human review

REUSABLE TEMPLATE

```

ROLE: [stable role]
OBJECTIVE: [stable objective]
RULES:
- [rule 1]
- [rule 2]

INPUTS:
<source>
{{SOURCE_TEXT}}
</source>

OUTPUT:
[stable schema or structure]

CHECK:
Before answering, verify the response satisfies every rule above.
```

Stable vs variable parts

Stable instructions

- Purpose
- Definitions
- Output schema
- Quality rules
- Safety constraints

Variable inputs

- User question
- Source text
- Job description
- Dataset
- Target audience

WORKFLOW EXAMPLE

Resume workflow: extract facts -> identify target-role requirements -> draft bullets -> check for unsupported claims -> revise -> human approval. Each step has a narrower prompt and a clearer

evaluation target.

CHAPTER 10

Safety, Privacy, and Professional Use

Good prompting includes knowing what should not be delegated or exposed.

Learning objectives

- Avoid unnecessary sensitive data
- Recognize hallucination and overconfidence risks
- Use citations or source grounding when needed
- Keep humans accountable for consequential decisions
- Do not paste confidential company data into tools that are not approved for it.
- Minimize personal data: include only what the task genuinely needs.
- Ask the model to distinguish facts, assumptions, and recommendations.
- For sourced work, require citations to the provided material and verify them.
- Use human review for hiring, legal, financial, medical, security, and other high-impact decisions.

GROUNDING PROMPT

Use only the information inside <source>.
If the source does not support an answer, say "Not supported by the provided source."
Separate direct source facts from your interpretation.
Do not fill gaps from general knowledge.
<source>
{{SOURCE}}
</source>

COMMON MISTAKE

Treating confident wording as evidence. Fluency is not verification. If correctness matters, build a verification step into the workflow.

CAPSTONE PROJECT

CareerSense Job-Search Copilot

Build a small prompt system that helps a candidate analyze a job description, tailor resume evidence, prepare interview stories, and create an application checklist without inventing experience.

Project goal

The copilot accepts two inputs: a candidate fact sheet and a target job description. It produces four outputs: role requirements, evidence mapping, interview preparation, and a final application checklist.

CORE RULE

The system must never fabricate experience, certifications, dates, metrics, or tools. When evidence is missing, it should label the gap explicitly.

1. Extract requirements from the job description.
2. Map each requirement to evidence from the candidate facts.
3. Identify unsupported gaps without inventing experience.
4. Draft resume bullet options only from supported facts.
5. Create interview questions and select relevant evidence.
6. Produce an application-readiness checklist.

CAPSTONE - PROMPT 1

Requirement Extractor

PROMPT

You are analyzing a job description for a candidate.
Extract only requirements explicitly present in the job description.
Group them into: Core Responsibilities, Technical Skills, Business Skills, Experience Signals, and Nice-to-Haves.
For each item include a short evidence quote from the job description.
Do not infer hidden requirements.
Return a markdown table.

CAPSTONE - PROMPT 2

Evidence Mapper

PROMPT

Compare the candidate fact sheet with the extracted requirements.
For each requirement assign one status: Strong Evidence, Partial Evidence, No Evidence.
Quote the exact candidate fact supporting the status.
Never create new experience.
For No Evidence, suggest a truthful preparation action rather than a fake resume claim.

CAPSTONE - PROMPT 3

Interview Planner

PROMPT

Create 8 interview questions from the target role.
For each question, recommend which verified candidate story or fact should be used.
If no relevant evidence exists, mark it as a preparation gap.
Include one follow-up question per item.
Keep questions realistic and role-specific.

CAPSTONE - QUALITY CHECK

Application Readiness Review

Check	Pass condition
Truthfulness	Every claim maps to candidate evidence.
Relevance	Output reflects the target job description.
ATS clarity	Keywords are used naturally, not stuffed.
Interview readiness	Key responsibilities have a prepared example.
Gaps	Missing evidence is clearly identified.
Actionability	Next steps are specific and realistic.

FINAL CHECKER

Review the application package against the six checks above.
 Return PASS or NEEDS WORK for each.
 For NEEDS WORK, give one specific fix.
 Do not add new candidate facts.
 Finish with the top 3 actions to complete before applying.

CAREERSENSE PRINCIPLE

AI should make career preparation more structured and personalized, not less truthful. Use prompts to surface evidence and gaps - never to manufacture credentials.

QUICK REVISION

Prompt Engineering - One-Page Mental Model

- Start with a precise task: use a verb and a concrete deliverable.
- Add only context that can change the answer.
- Use constraints to remove ambiguity and prevent unwanted behavior.
- Specify output format when consistency matters.
- Use examples when style, labels, or boundaries are hard to describe.
- Decompose complex work into smaller stages.
- Separate generation from evaluation.
- Turn successful prompts into reusable templates.
- Test prompts against multiple realistic inputs.
- Ground high-stakes outputs in trusted sources and human review.

SIMPLE FORMULA

TASK + CONTEXT + CONSTRAINTS + OUTPUT + QUALITY BAR = a strong default prompt structure.

REFERENCE

15 High-Value Prompt Patterns

Pattern	Template
Rewrite	Rewrite X for audience Y while preserving facts.
Extract	Extract only A, B, and C from the source.
Classify	Assign exactly one label from the allowed set.
Compare	Compare options against explicit criteria.
Summarize	Summarize for a specific audience and decision.
Critique	Evaluate against a rubric and identify weak sections.
Revise	Fix only the issues identified in the critique.
Transform	Convert source format A into format B.
Brainstorm	Generate options within clear constraints.
Plan	Create steps, dependencies, risks, and checkpoints.
Analyze	Separate observations, hypotheses, and actions.
Ground	Use only supplied sources; say when unsupported.
Code	Specify environment, inputs, outputs, and tests.
Tutor	Explain, ask a check question, then adapt.
Simulate	Act as a role while obeying explicit task rules.

TROUBLESHOOTING

10 Common Prompting Mistakes

1. Vague verbs: “improve,” “analyze,” or “make better” without defining the deliverable.
2. Too much irrelevant context that distracts from the real task.
3. Conflicting instructions, such as “be comprehensive” and “use only 50 words.”
4. Requesting a format without giving a schema or example.
5. Using personas as a substitute for facts or domain validation.
6. Asking for unsupported certainty when the data is incomplete.
7. Copy-pasting generated content without verification.
8. Changing many prompt variables at once during testing.
9. Using examples that accidentally teach the wrong pattern.
10. Sending sensitive information that is not necessary for the task.

PRACTICE

Practice Questions

1. What five elements make up the default prompt structure used in this handbook?
2. Why is “Act as an expert” not enough to make a prompt reliable?
3. When should you prefer a structured JSON output?
4. What is the purpose of few-shot examples?
5. Why can contradictory examples override a written instruction?
6. What does task decomposition improve?
7. How should you separate observations from hypotheses in analysis?
8. What is the difference between stable prompt instructions and variable inputs?
9. Why should generation and evaluation often be separate steps?
10. What does grounded prompting mean?
11. Name three pieces of information that improve a coding prompt.
12. Why should a resume prompt forbid invented metrics?
13. What is a prompt test set?
14. Why is changing one major variable at a time useful?
15. When is human review especially important?

HANDS-ON

Hands-on Exercises

1. Repair a vague prompt

Turn “Make my resume better” into a prompt with task, context, constraints, output, and quality bar.

2. Build a summarizer

Create a prompt that summarizes a long report for an executive in exactly five bullets.

3. Create a classifier

Design four support-ticket labels and provide three representative examples.

4. Design a JSON extractor

Extract name, company, role, date, and status from job-search notes.

5. Create a critique loop

Draft an email, evaluate it against a rubric, then revise only weak sections.

6. Build a grounded Q&A prompt

Require answers to come only from a supplied policy document.

7. Write a code prompt

Specify a small data-cleaning task including version, libraries, input, output, and test.

8. Create a reusable template

Separate stable instructions from a variable job description.

9. Build a mini test set

Create five different inputs that could expose failures in your prompt.

10. Run the capstone

Use a real job description and a fact sheet you control to test the Job-Search Copilot workflow.

REFERENCE

Glossary

Term	Plain-English meaning
Prompt	The instructions and context given to a generative model.
Context	Information that helps the model interpret the task.
Constraint	A rule that limits or shapes the output.
Output schema	A defined structure the response must follow.
Few-shot prompting	Providing examples that demonstrate the desired pattern.
Zero-shot prompting	Requesting a task without worked examples.
Grounding	Restricting answers to trusted supplied information.
Hallucination	A plausible-sounding output that is unsupported or incorrect.
Evaluation rubric	Criteria used to judge response quality.
Prompt template	A reusable prompt with stable instructions and variable fields.
Decomposition	Breaking a complex task into smaller stages.
System instruction	Higher-level behavior or rules applied to the model.
User instruction	The current task requested by the user.
Structured output	A response following a predictable schema such as JSON.
Test set	A collection of representative inputs used to evaluate a prompt.
Human-in-the-loop	A workflow where a person reviews or approves important output.
Iteration	Improving a prompt through controlled testing and revision.

Retrieval grounding	Supplying relevant source material for the model to use.
Temperature	A generation setting that can influence randomness where supported.
Token	A unit of text processed by the model.

FINAL ASSESSMENT

20-Question Quiz

1. Which prompt element defines what the model should do?

A. Context B. Task C. Example D. Persona

2. What is the best reason to add context?

A. Make prompt longer B. Change answer meaningfully C. Sound expert D. Add decoration

3. Which is best for machine-readable output?

A. Vague prose B. JSON schema C. Long persona D. More adjectives

4. Few-shot prompting primarily uses what?

A. Tools B. Examples C. Images D. Citations

5. What should you do when source evidence is missing?

A. Guess B. Invent C. Mark unsupported D. Ignore

6. Why decompose complex tasks?

A. Add length B. Reduce ambiguity C. Hide errors D. Avoid context

7. A good code prompt should specify what?

A. Only language B. Environment, inputs, outputs, constraints C. Persona only D. Nothing

8. What is grounding?

A. Writing casually B. Restricting output to trusted sources C. Removing format D. Making output longer

9. Why use an evaluation rubric?

A. Judge quality consistently B. Increase randomness C. Replace sources D. Add personality

10. What should remain stable in a reusable prompt template?

A. Variable data B. Core rules and schema C. User name D. Every input

11. What is a hallucination?

A. Verified fact B. Unsupported generated claim C. Table format D. Prompt template

12. When should a persona be used?

A. To replace facts B. To focus perspective C. To guarantee expertise D. To avoid task instructions

13. What is the safest treatment of sensitive data?

A. Include everything B. Minimize and use approved tools C. Publish it D. Ignore policy

14. What does a test set help evaluate?

A. Prompt consistency B. Font choice C. File size D. Internet speed

15. Which workflow is often stronger?

A. Generate only B. Generate-check-improve C. Repeat same prompt D. Remove criteria

16. What does a constraint do?

A. Narrow acceptable output B. Add unrelated history C. Replace task D. Create facts

17. Why specify audience?

A. Control relevance and language B. Increase token count C. Avoid structure D. Guarantee correctness

18. What should a resume prompt never invent?

A. Formatting B. Candidate facts and metrics C. Headings D. Bullet order

19. Why change one major prompt variable at a time?

A. Identify what caused improvement B. Make testing slower C. Avoid evaluation D. Increase randomness

20. Who remains accountable for consequential decisions?

A. The model B. The prompt C. The human organization/user D. Nobody

ANSWER KEY

Quiz Answers

Questions 1-10	Questions 11-20
1-B, 2-B, 3-B, 4-B, 5-C, 6-B, 7-B, 8-B, 9-A, 10-B	11-B, 12-B, 13-B, 14-A, 15-B, 16-A, 17-A, 18-B, 19-A, 20-C

If you missed a question, return to the chapter that introduced the concept and rewrite one example prompt yourself. Active reconstruction is more useful than rereading the answer key.

CLOSING THE BOOK

Prompting Is a Professional Communication Skill

The strongest prompt engineers are not people who know secret phrases. They are people who can define a task, identify the information that matters, set boundaries, evaluate quality, and improve a workflow over time.

- Be specific about the deliverable.
- Be selective about context.
- Be explicit about constraints.
- Be disciplined about verification.
- Be honest about uncertainty.
- Keep humans responsible for important decisions.

NEXT STEP

Choose one recurring task from your work this week. Turn it into a reusable prompt template, test it on five inputs, record the failures, and improve it once. That is prompt engineering in practice.

CareerSense

Build skills. Prove your value. Grow your career.

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